# RoboShop Ansible Server SSH Setup Guide

Complete process from copying the AWS PEM key from Windows PowerShell to successfully pinging all RoboShop target hosts with Ansible.

**Environment:** Ansible server running as root; target EC2 servers use ec2-user.

## 1. Copy the AWS PEM Key from Windows to the Ansible Server

The AWS private key was stored on the Windows PC. Copy it to the Ansible server using SCP. Use the Ansible server public IP from Windows.

```
# From Windows PowerShell
scp -i "C:\path\to\AnsibleServer.pem" "C:\path\to\DevOps_AWS.pem" ec2-user@<ANSIBLE-SERVER-PUBLIC-IP>:/home/e

# Example
scp -i "C:\path\to\AnsibleServer.pem" "C:\Users\pavan\Downloads\DevOps_AWS.pem" ec2-user@54.242.130.134:/home
```

## 2. Become Root on the Ansible Server

The Ansible server is being used as root.

```
ssh -i "C:\path\to\AnsibleServer.pem" ec2-user@<ANSIBLE-SERVER-PUBLIC-IP>
sudo -i
```

## 3. Secure the PEM Key

Move the PEM to root's home if required, then set root ownership and restrictive permissions.

```
# If the file is in /home/ec2-user/
mv /home/ec2-user/DevOps_AWS.pem /root/

chown root:root /root/DevOps_AWS.pem
chmod 400 /root/DevOps_AWS.pem

ls -l /root/DevOps_AWS.pem
```

## 4. Test the PEM from the Ansible Server

Confirm the existing AWS key can reach a RoboShop server.

```
ssh -i /root/DevOps_AWS.pem ec2-user@catalogue.roboshopservice.store

# After successful login
exit
```

## 5. Generate a Dedicated SSH Key for Ansible

Generate the Ansible server's own SSH key pair as root.

```
ssh-keygen -t rsa

# Press Enter for the default:
# /root/.ssh/id_rsa

# Leave the passphrase empty for an automation key.
```

## 6. Verify the New SSH Key

The private key remains on the Ansible server. Only the public key is copied to target servers.

```
ls -l /root/.ssh/id_rsa*

# Private key:
# /root/.ssh/id_rsa

# Public key:
# /root/.ssh/id_rsa.pub
```

## 7. Bootstrap the Public Key to All RoboShop Servers

Use the original AWS PEM only for this initial bootstrap. The loop adds the Ansible public key to each ec2-user authorized_keys file.

```
for host in catalogue mongodb user cart redis shipping mysql rabbitmq payment web
do
    echo "Configuring $host..."

    ssh -i /root/DevOps_AWS.pem \
        -o StrictHostKeyChecking=no \
        ec2-user@$host.roboshopservice.store \
        "mkdir -p ~/.ssh && chmod 700 ~/.ssh && echo '$(cat /root/.ssh/id_rsa.pub)' >> ~/.ssh/authorized_keys
done
```

## 8. Test SSH Without the PEM

After bootstrapping, the Ansible server should connect using its own id_rsa key.

```
ssh ec2-user@catalogue.roboshopservice.store
exit

ssh ec2-user@mongodb.roboshopservice.store
exit
```

## 9. Configure the Ansible Inventory

Use the Route53 hostnames and specify ec2-user plus the new Ansible private key.

```
[catalogue]
catalogue.roboshopservice.store

[mongodb]
mongodb.roboshopservice.store

[user]
user.roboshopservice.store

[cart]
cart.roboshopservice.store

[redis]
redis.roboshopservice.store

[shipping]
shipping.roboshopservice.store

[mysql]
mysql.roboshopservice.store

[rabbitmq]
rabbitmq.roboshopservice.store

[payment]
payment.roboshopservice.store
```

```
[web]
web.roboshopservice.store

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=/root/.ssh/id_rsa
```

## 10. Install Ansible Core if Required

If the ansible command is not found on CentOS 9, install Ansible Core.

```
dnf install epel-release -y
dnf install ansible-core -y

ansible --version
```

## 11. Run the Ansible Ping Test

From the RoboShop Ansible project directory, use the inventory file and Ansible ping module.

```
cd /root/Ansible-roboshop

ansible all -i inventory -m ping
```

## 12. Expected Result

All target hosts should return SUCCESS and ping: pong. This confirms DNS, SSH authentication, Python and Ansible connectivity.

```
cart.roboshopservice.store | SUCCESS
catalogue.roboshopservice.store | SUCCESS
redis.roboshopservice.store | SUCCESS
mongodb.roboshopservice.store | SUCCESS
user.roboshopservice.store | SUCCESS
shipping.roboshopservice.store | SUCCESS
mysql.roboshopservice.store | SUCCESS
rabbitmq.roboshopservice.store | SUCCESS
payment.roboshopservice.store | SUCCESS
web.roboshopservice.store | SUCCESS

# Each successful host should contain:
# "ping": "pong"
```

## Final Architecture

```
Windows PC
   |
   | AWS PEM key
   v
Ansible Server (root)
   |
   +-- /root/DevOps_AWS.pem       (bootstrap/emergency key)
   |
   +-- /root/.ssh/id_rsa          (Ansible private key)
   +-- /root/.ssh/id_rsa.pub      (Ansible public key)
                |
                +--> catalogue
                +--> mongodb
                +--> user
                +--> cart
                +--> redis
                +--> shipping
```

```
        +--> mysql
        +--> rabbitmq
        +--> payment
        +--> web
```

## Important Notes

Never copy or share /root/.ssh/id_rsa. Only id_rsa.pub is placed in target hosts' authorized_keys. The initial AWS PEM is used to bootstrap the Ansible key. Target servers use ec2-user for SSH. Port 22 must be reachable from the Ansible server. In the current AWS setup, web uses a public Route53 IP while the other services use private IPs.

## Useful Verification Commands

```
whoami
ansible --version
ansible-inventory -i inventory --graph
ansible all -i inventory -m ping
ansible catalogue -i inventory -m ping
ansible all -i inventory -a "hostname"
```